



Hedging & Financing

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What Is Hedging?

DEFINITION

A **hedge** is a position taken to offset the price risk of something you already own or need. When prices move, the gain on one side pays for the loss on the other.

WHAT HEDGING DOES

- **Reduces risk**
- **Makes the future predictable**

WHAT HEDGING DOESN'T DO

- **Make you money**
- **Remove all risk**

Who Hedges?

	Short hedge	Long hedge
Who	Producers: farmers, miners, oil producers	Consumers: airlines, food makers, refiners
Physical position	Own the commodity, will sell later	Need the commodity, will buy later
Afraid of	Falling prices	Rising prices
The hedge	Sell futures today	Buy futures today
Example	Corn farmer sells December corn futures	Airline buys crude or heating oil futures

Real-world hedgers

Mexican government

Ryanair

Worked Example

THE SETUP

- 01 Spring: expects 100,000 bushels at harvest
- 02 Sells 20 December corn futures at \$4.50 (5,000 bushels each)
- 03 Harvest: sells the corn locally, buys the futures back

THE OUTCOME

	Falls to \$3.80	Rises to \$5.20
Corn sold	\$380,000	\$520,000
Futures P&L	+\$70,000	-\$70,000
Total	\$450,000	\$450,000

UNHEDGED

\$380k–\$520k

but

HEDGED

\$450k

**The price is locked,
whichever way the
market moves.**

Futures vs Options

SAME FARMER, TWO HEDGES

Alternatively, buy a \$4.50 put. Assume a \$0.25/bu premium, purely for illustration.

Per bushel	Upfront cost	Corn falls to \$3.80	Corn rises to \$5.20	What you get
Unhedged	None	\$3.80	\$5.20	Full exposure
Futures hedge	None (margin only)	\$4.50	\$4.50	Price locked
Put option	\$0.25 premium	\$4.25	\$4.95	Floor + upside

FUTURES HEDGE

Free to put on and gives total certainty, but the farmer hands back every dollar of upside.

PUT OPTION

Costs the premium upfront, but sets a floor and keeps most of the upside if prices rise.

Hedge Ratios

MINIMUM-VARIANCE HEDGE RATIO

$$h^* = \rho \times \sigma_S \div \sigma_F$$

How many futures to use per unit of exposure

→ ρ : correlation of spot and futures price changes

→ σ_S, σ_F : volatility of each

→ Contracts: $N^* = h^* \times \text{exposure} \div \text{contract size}$

EXAMPLE

Airline hedging jet fuel with heating oil futures

$$h^* = 0.90 \times 2.5\% \div 3.0\% = \mathbf{0.75}$$

$$0.75 \times 2\text{m gal} \div 42,000 \text{ gal} = \mathbf{36 \text{ contracts}}$$

DELTA HEDGING

$$\Delta = \partial V \div \partial S$$

How much an option moves per \$1 move in price

→ Calls: **0 to 1**. Puts: **-1 to 0**

→ Hedge by holding the **opposite delta** in futures

→ Delta shifts as prices move: **rebalance constantly**

EXAMPLE

Bank that sold the farmer's \$4.50 put

$\Delta = -0.50 \rightarrow$ sells **10 corn futures**

Corn falls, $\Delta = -0.80 \rightarrow$ sells **6 more**

Hedging in the Real World

WHEN IT WORKS

Ryanair, 2026

Hedged before the Iran war

- Locked in **80%** of its fuel at **\$67/bbl**
- Then jet fuel hit **\$150/bbl**
- Less-hedged rivals are **raising fares and cutting flights**

WHEN IT DOESN'T

Metallgesellschaft, 1993

Sold 5–10 year fixed-price oil

- Hedged with **rolling short-dated futures**
- Oil fell: **margin calls** needed cash immediately
- Unwound at a loss of **~\$1.3bn**

THE BRIDGE TO FINANCING

A hedge needs cash behind it → that is where financing comes in



Financing Commodity Trading

The Big Question: Where Does the Money Come From?

ONE OIL CARGO (ILLUSTRATION)

CARGO SIZE	1 million barrels
PRICE	\$80 per barrel
CARGO VALUE	\$80 million
TRADER'S MARGIN	About \$1 per barrel
TRADER'S PROFIT	About \$1 million

TODAY'S ROADMAP

- 01 Why traders need to borrow**
The cash gap between paying and getting paid
- 02 How they borrow**
Loans tied to one cargo, and loans to the whole company
- 03 Why lenders say yes**
Hedging, and what has changed since 2020

Thin margins + huge cargoes = a business that runs on borrowed money

If the trader does not have \$80 million sitting in the bank, who pays the seller?

What Commodity Traders Do:

Move Goods Across Space, Time & Form



IN PLAIN ENGLISH

Commodity traders are essentially logistics companies. They buy raw materials, move or change them, and sell them, using financial markets to pay for it all and to limit price risk.

SPACE

Ship it

Move goods from where they are produced to where they are used.

Example: crude oil from West Africa to a refinery in Europe.

TIME

Store it

Hold goods when there is too much and sell them when demand picks up.

Example: keep heating oil in tanks until winter.

FORM

Blend it

Mix different grades so the product matches what a customer needs.

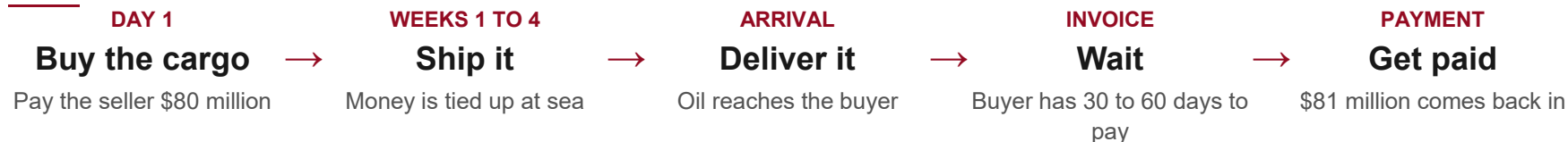
Example: blend two fuels to meet one buyer's quality standard.

Every step takes weeks, and the goods must be paid for the whole time.

Large trading firms include Vitol, Trafigura, Glencore, Gunvor and Cargill.

Why Financing Is the Core Problem: The Cash Gap

FOLLOW ONE \$80 MILLION CARGO (ILLUSTRATION)



The cash gap: about 2 to 3 months with \$80 million tied up

IN PLAIN ENGLISH

Working capital: the cash a business needs to cover the gap between paying for goods and getting paid for them.

Collateral: something valuable a lender can take and sell if a loan is not repaid.

THE KEY FACT

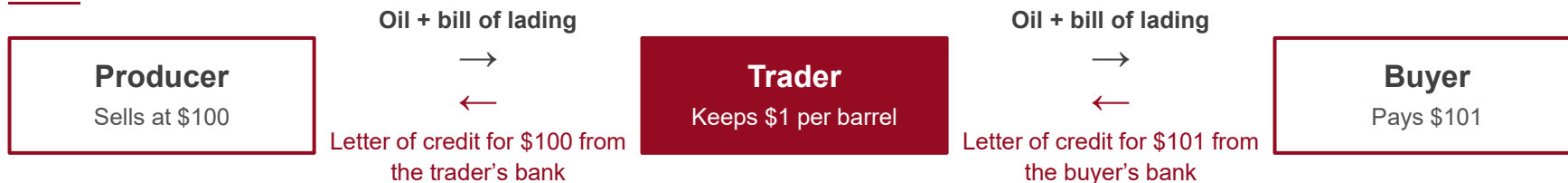
Traders buy and sell far more than their own money could cover. Banks lend them the rest because the loans are backed by the commodities themselves.

Letters of Credit: A Bank's Promise to Pay

DEFINITION

A **letter of credit** is a document from a bank promising the seller will be paid in full, as long as the agreed delivery conditions are met.

HOW ONE TRADE WORKS



Nobody has to trust a stranger: each side relies on a bank's promise.

The bank is repaid when the cargo is sold, so the loan pays itself off.

Loans Tied to One Cargo:

Four Transaction Finance Tools

EACH LOAN IS REPAID FROM THE SALE OF THE CARGO IT PAID FOR

Tool	How it works, in plain English	When it is used
Letter of credit	A bank promises the seller will be paid once the shipping documents are shown.	Buying the cargo
Inventory financing (a repurchase agreement, or “repo”)	The trader sells the stored goods to a bank and agrees to buy them back later at a set price. It works like a loan backed by the goods.	While goods sit in storage
Receivables financing	The trader sells the buyer’s unpaid invoice (a “receivable”) to a bank or fund, and gets cash now instead of waiting.	After delivery, before the buyer pays
Prepayment financing	The trader pays a mine or oil producer upfront for future supply, often with money borrowed from a group of banks.	Locking in future supply

Company-Level Financing: Money for the Whole Firm



WHY CARGO LOANS ARE NOT ENOUGH

Cargo loans are short-term, and banks are not obliged to keep offering them. Traders also need backup cash for volatile markets and long-term money for things like ports and storage.

SIZED TO WHAT YOU OWN

Borrowing base

A credit line sized to everything the trader holds right now: stored goods plus money customers owe. It is updated regularly.

Example: \$150 million of goods and invoices lets the trader borrow a set share of it, say \$120 million (illustration).

REUSABLE CREDIT LINE

Revolving credit facility

A large credit line from a group of banks. The firm can borrow, repay and borrow again, like a company credit card.

Example: cash for daily futures margin calls when prices jump.

LONG-TERM MONEY

Bonds and securitization

Borrowing from investors for years instead of months. One version bundles customer invoices into a separate company that sells bonds.

Example: Trafigura issued a bond with no repayment date in 2013.

Cargo loans run the daily business. Company loans keep the firm alive in a crisis.

Why Banks Say Yes: Hedging Protects the Collateral



REMEMBER FUTURES FROM EARLIER?

When a trader buys a cargo, it usually sells oil futures at the same time. If oil falls, the cargo loses value, but the futures gain about the same amount.

WITHOUT A HEDGE

Cargo bought at \$80 per barrel	\$80 million
Oil falls to \$70 per barrel	\$70 million

-\$10 million

The bank's collateral just shrank.

WITH A HEDGE

Cargo value change	-\$10 million
Futures gain (sold at \$80)	+\$10 million

About \$0

Value is locked in, so the lender is protected.

Hedged cargo is safer collateral, which keeps lenders willing to lend

The catch: futures losses must be paid in cash every day (margin calls), so traders keep extra credit lines ready.

What Is Changing: Banks Pull Back, New Lenders Step In



WHAT HAPPENED

2020

Fraud cases

Oil prices crashed and several Singapore traders collapsed. Hin Leong owed about \$3.5 billion to 23 banks after faking sales documents and selling oil it had pledged as collateral.

2020

→ Banks pull back

ABN Amro quit commodity trade finance completely. BNP Paribas, Société Générale and Natixis also exited or froze new deals.

SINCE THEN

→ New lenders step in

Private credit funds (investment funds that lend directly) have filled part of the gap, especially for smaller traders.

THE LESSON

Collateral only protects a lender if it is real and pledged to one lender at a time.

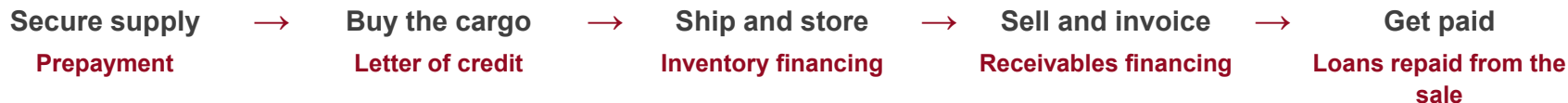
WHAT TO WATCH

Digital shipping documents and shared collateral records, built to stop the same cargo being pledged twice.

Sources: Reuters, Bloomberg and The Straits Times reporting (2020); TXF, "Private credit has a growing role in trade and commodity finance" (2021).

Putting It Together: One Cargo, Every Tool

FOLLOW THE CARGO



Underneath it all: hedging protects the cargo's value, and company credit lines provide backup cash.

THREE THINGS TO REMEMBER

- 01** Financing is the business model: traders move goods worth far more than their own capital.
- 02** Most loans are tied to a real cargo and are repaid when that cargo is sold.
- 03** Hedging and real, trustworthy collateral are why lenders say yes.